

Artificial Intelligence Assignment: 1

Submitted by:

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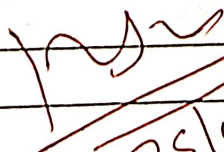
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Branch: B.Tech CSE with
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Semester: III

Submitted to:

Prof. Ritu Raj Jha


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① What is Artificial Intelligence?

→ AI is a branch of computer science that enables machines to mimic human intelligence - like learning, reasoning, and decision-making.

② List any five applications of AI.

→ Five applications are listed below:

- self-driving cars
- Chatbots in customer support
- Disease prediction in healthcare
- Fraud detection in finance
- Personalized learning in education.

③ Difference between strong and weak AI.

→ The difference are

Weak AI: Performs specific tasks and have not much intelligence like human's ex: Alexa, Siri.

Strong AI: Thinks, learn, and reason like human.

They are not yet discovered or developed.

④ What is the role of intelligence agent in AI?

→ They observe their environment, process data, and take actions to achieve goals automatically.

⑤ Explain the term "AI techniques" with examples.

→ An AI techniques is a method used to solve problems.
Ex: Search algorithms, knowledge representation, ML, NLP, Computer vision.

⑥ State the three limitations of AI.

→ They are listed below:-

- High development and maintenance cost.
- No emotional understanding
- Lacks true creativity and intuition.

⑦ Briefly describe the AI Ladder.

→ IBM proposed a ladder to help companies adopt AI:

- i) Collect - Gather relevant data
- ii) Organize - Clean and label the data
- iii) Analyze - Apply AI/ML to gain insights
- iv) Infuse - Integrate AI into business processes
- v) Modernize - Update continuously for improvement.

⑧ What are the criteria of success in AI?

→ They are:

- Accuracy in tasks
- Efficiency in processing
- Adaptability to new data
- Usability for easy human interaction.

Write a short note on learning agents.

→ They improve over time by learning from feedback, past experiences, and trial-and-error interactions.

(10) Describe the impact of AI on the healthcare industry.

→ AI helps detect diseases early, enables robotic surgery, and supports faster, more accurate diagnoses.

(11) Explain utility-based agents.

→ They choose actions based on a utility function, aiming for the most beneficial and effective outcome.

(12) How does AI differ from Machine Learning?

→ AI is broader concept of smart machines which can replicate human behaviour and knowledge. ML is subset that enables machines to learn from data patterns.

(13) What do you mean by rational agents?

→ Rational agents choose the best action based on their knowledge and goals to maximize performance.

Q4) Mention five career options in AI.

→ The career options in AI are as:

- Machine Learning Engineer
- Data Scientist
- AI Researcher
- Robotics Engineer
- NLP Engineer

Q5) Describe the importance of perception in AI.

→ Perception lets AI systems sense and interpret their environment using tools like cameras, sensors, and microphones for better decision-making.

Artificial Intelligence Assignment: 1B

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Define Artificial Intelligence. Explain its goals and scopes in modern computing.

→ Tech that makes machines acts smart, like humans.

Goals:

- Solve problems
- Learn from data
- make decision like humans.

Scopes in modern computing:

- Healthcare (disease prediction)
- Transport (self-driving cars)
- Finance (fraud detection)
- Education (personalized learning)
- Retail, Agriculture - It's literally everywhere.

② Explain the architecture of an Intelligent agent. Describe various types of agents with suitable examples.

→ Architecture:

- Sensors: collect data (eg, camera, mic)
- Actuators: take action (eg, wheels, arms)
- Percepts: what it senses
- Performance measure - How well it performs.

Types of Agents:

- Simple reflex: Reacts to now (e.g smart lights)
- Model-Based: Has memory (e.g robo vacuum)

- Goal-Based: Picks actions to reach a goal (e.g. chess bot)
- Utility Based: Chooses best outcomes (e.g. shopping bot).
- Learning Agent: Learns and improves (e.g. Youtube).

③ Differentiate between strong AI and weak AI. Discuss the applications and limitations of each type.

→ Weak AI (Narrow)

- Specific task only
- Real (e.g. Alexa, Siri)
- Can't feel or understand
- Easy to build

Strong AI (General)

- Human-level understanding
- Skill: theoretical
- Could understand emotions
- Super hard + complex

④ Describe Learning agents. Explain its components and how it improves performance over time with a diagram.

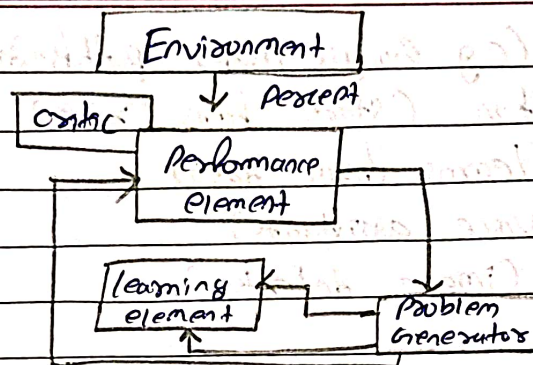
→ Agents that improve performance by learning from past experiences.

Components

- Learning element - Learns from data
- Performance element - makes decisions
- Critic - gives feedback
- Problem Generator - suggest new tasks

How it improves

Tries → Gets feedback → Adjusts → Tries again.



⑤ Explain the various applications of AI in real-world domains like healthcare, education and robotics.

→ The various application of AI in real-world domains like healthcare, education and robotics are as follows:

- Healthcare : Robot-assisted surgeries which help people to get treatment on time. Disease prediction with proper analysis like detection, classification.
- Education : AI tutors which can adapt the student's study pattern, so that personalized learning plans can be developed.
- Robotics : Factory automation by which manufacturing will grow up fast, home cleaning robot decreases human efforts etc.

⑥ Describe the AI techniques with suitable examples. How are AI techniques used in intelligent decision-making?

→ Techniques of AI are:

- Search Algorithms (e.g. Google maps pathfinding)
- Knowledge representation (store facts / rules)
- Machine Learning (learns from data)
- NLP (chatbots, voice assistants)
- Computer vision (image detection).

Use in Decision-making:

→ AI uses data + logic to make smart choices - like diagnosing a patient, recommending a product, or navigating traffic.

⑦ What is a rational agent? How is it different from a reflex agent? Explain with real-life examples.

→ Rational Agent: Thinks before acting, picks best action. e.g. chess AI

Example: You planning a trip with budget + weather in mind.

Reflex agent: Instant reaction, no deep thinking
e.g. auto door sensors

Example: You pulling your hand back from a hot pan

⑧ Compare Narrow AI, General AI, and Super AI. Provide Examples and discuss their current development status.

→ Comparison is as follows

TYPE	Description	Example	Status
N. AI	task-specific AI	Siri, Netflix	Exists Now
G. AI	Human-level smart	Human like robo	in research
S. AI	Smarter than humans	sci-fi level stuff	Future only.

⑨ What are the major advantages and disadvantages of AI? Illustrate with examples.

→ The advantages and disadvantage of ~~are~~ AI are listed below:

Advantages

- Super fast + efficient
- No breaks needed
- Good at risky work (e.g. space, mines)

Disadvantages

- High cost
- No creativity or emotions
- Needs a lot of data
- can replace human jobs

Example: AI can fly planes, but it won't understand a passenger's panic during turbulence

⑩ Describe the structure and working of utility-based agents. How are they different from goal-based agents?

→ Utility-Based Agents:

- tries to pick the best action using a utility function.
- Ex: AI in self-driving cars choosing safest + fastest route

Goal-Based Agents:

- Focuses only on reaching a goal
- Ex: AI playing chess to win the game

key difference:

- Utility agents care about how good the result is.
- Goal agents just want to get there, even if it's not optimal

12/10/25